

Выпуклые множества:

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The three theorems of this sheet are the foundation of all convex geometry, and all three have elementary proofs. From here on they work everywhere: from olympiad problems about coverings to theorems on sections of convex bodies.

Definition 1. The **convex hull** $\text{conv } S$ of $S \subseteq \mathbb{R}^n$ is the set of all finite convex combinations of points of S ; equivalently, the smallest convex set containing S .

Theorem 1 (Carathéodory 1907). If $x \in \text{conv } S$ for $S \subseteq \mathbb{R}^n$, then x is a convex combination of at most $n + 1$ points of S .

Theorem 2 (Radon 1921). Any $n + 2$ points of $\mathbb{R}^n$ can be split into two disjoint groups whose convex hulls intersect.

Theorem 3 (Helly 1913, 1923). If among convex sets $K_1, \dots, K_m \subseteq \mathbb{R}^n$ ($m \geq n + 1$) any $n + 1$ have a common point, then all of them do. The same holds for infinite families of *compact* convex sets.

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1. (Carathéodory 1907) Prove Theorem 1 and show that $n + 1$ cannot be replaced by n . Deduce that the convex hull of a compact set in $\mathbb{R}^n$ is compact, and give an example of a closed set whose convex hull is not closed.
 2. (Radon 1921) Prove Theorem 2, and deduce the finite case of Helly's theorem by induction on the number of sets.
 3. (Helly 1923) Deduce the infinite version of Theorem 3 for compact convex sets. Show by examples that neither compactness nor the number $n + 1$ can be weakened.

4. (Krasnosel'skii 1946) Let $S \subseteq \mathbb{R}^2$ be a compact set such that for any three points of S there is a point of S seeing all three (i.e. joined to each of them by a segment inside S). Prove that S is star-shaped. (*Hint: for each $x \in \partial S$ consider the half-plane of visibility and apply Helly.*)
5. (ФКН, вечна 2021) Given a positive integer n and $2021n$ convex compact sets in $\mathbb{R}^n$ whose intersection can be covered by a strip of width 1 (a strip is the part of space between two parallel hyperplanes). Prove that one can choose $2n$ of these sets whose intersection can also be covered by a strip of width 1.
6. Prove the separation theorem: a point outside a closed convex set $K \subseteq \mathbb{R}^n$ can be strictly separated from K by a hyperplane, and two disjoint compact convex sets can be strictly separated. Give an example of two disjoint closed convex sets that cannot be strictly separated.
7. (classical) A finite set of points in the plane is such that every three of them can be covered by a disc of radius 1. Prove that all of them can be covered by a disc of radius 1. (*Hint: apply Helly to the unit discs centred at these points.*)